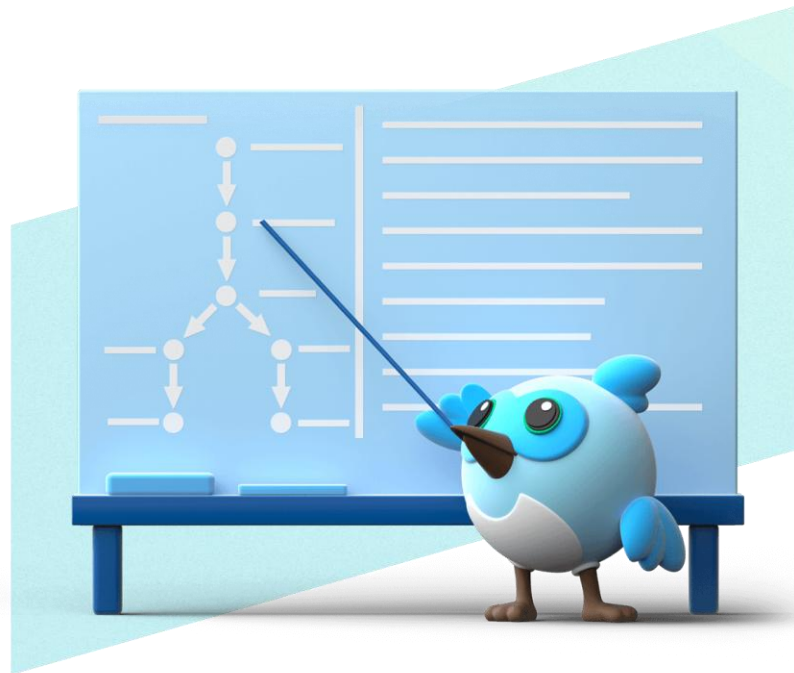


Let's the game **BEGIN**



WELCOME **GEN11** TO

MOBILE DEVELOPMENT FOUNDATIONS



CADT
IDT

RONAN OGOR
CADT Lecturer Researcher

THEA CHOEM
Mobile Software Engineer



MOBILE DEVELOPMENT FOUNDATIONS

KICK-OFF

About Mobile Dev

- What is **Flutter** exactly ?
- **Why** do you need this course?

Course organization

- Course **Learning objectives**
- Course **Agenda**
- Course **Pedagogy**
- Course **Grading**

Let's start with Dart !

- Variables
- Types (built in)
- Loops
- Functions



Is Flutter worth learning?

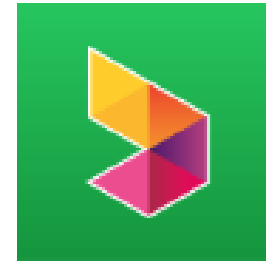
Just look at your **mobile apps** right now !



↑
Millions of real-time
transactions every
single day



↑
The job market is aggressively hunting for
engineers who know how to build Flutter apps

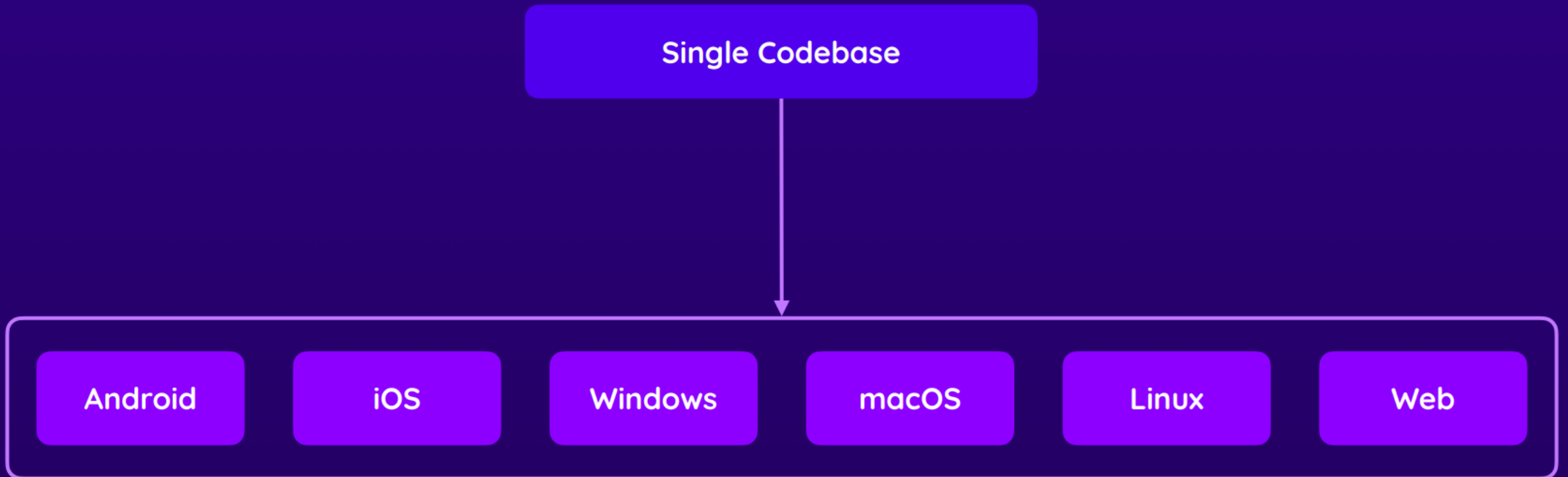


MORE CAMBODIAN APP

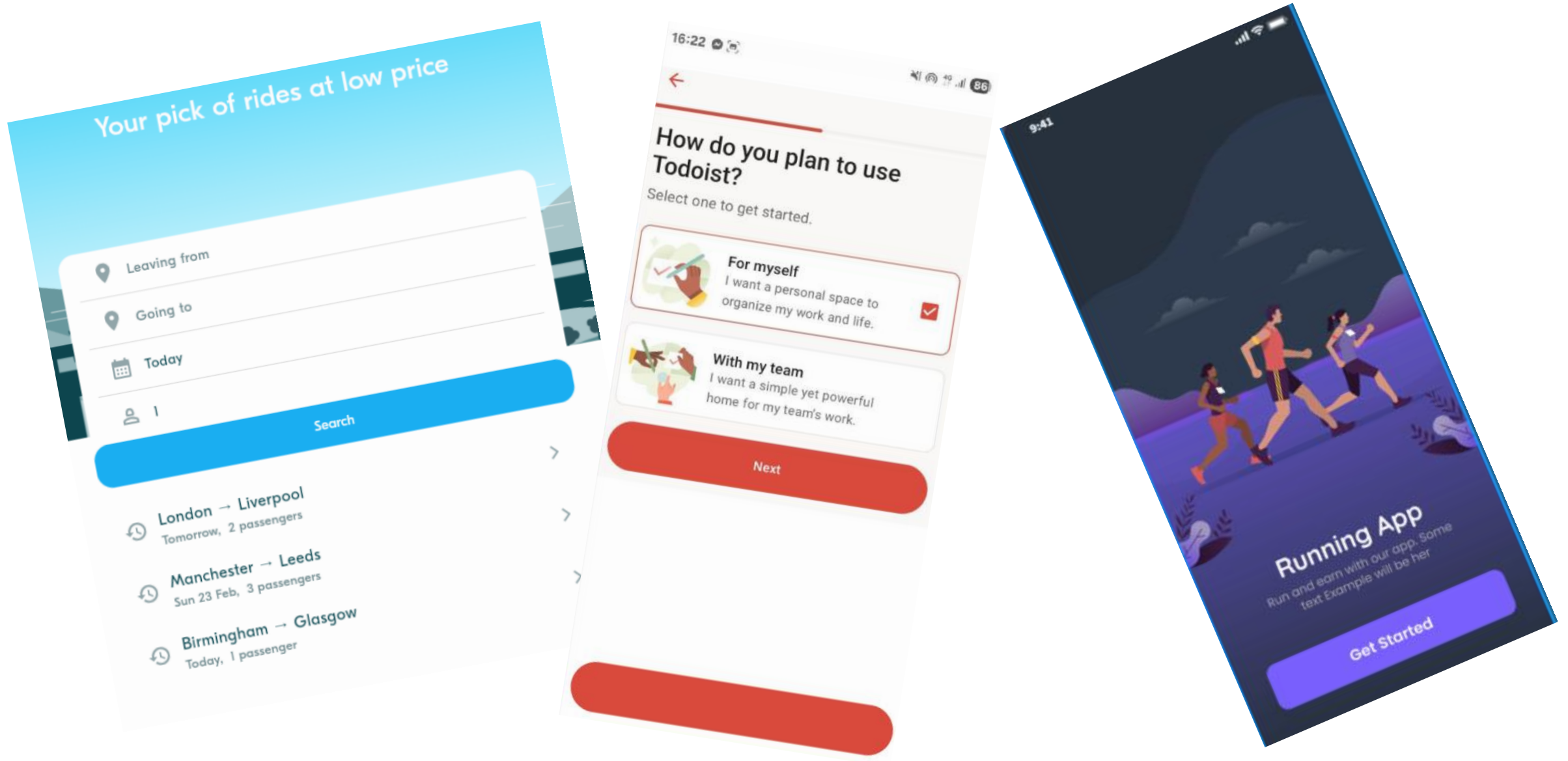


↑
Billions of real-time
transactions every
single day

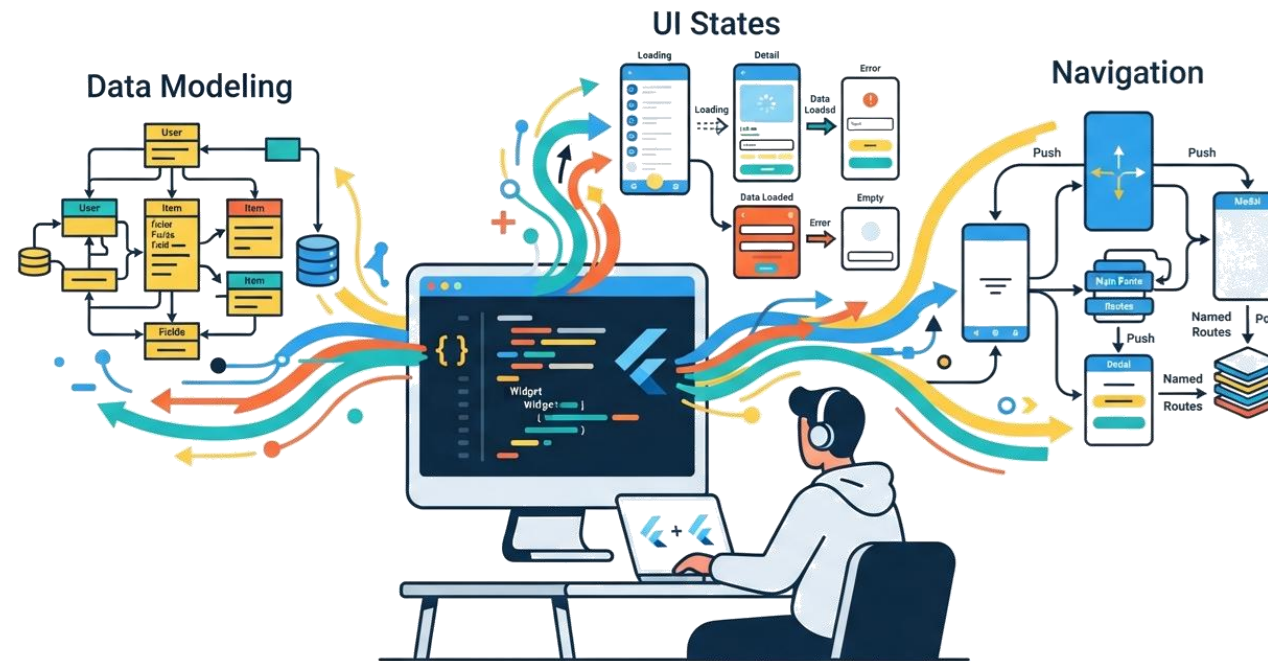
Flutter : a **cross-platform** development framework



A few Apps you will work on



COURSE ORGANIZATION



*Theoretical or information you **need to understand***



KNOW

- ✓ Core **Dart** and **OOP** concepts
- ✓ **Layered** architecture and **component** responsibilities
- ✓ Flutter **Widget tree**
- ✓ **Layout** & constraint mechanics
- ✓ **Stateless** vs **stateful** widgets
- ✓ **State** flow: setState, callbacks, lifting state up
- ✓ **Data** flow and **navigation**

*Practical abilities you **need to acquire***



SKILLS

- ✓ **Model domain** logic in Dart (OOP)
- ✓ Build a **layered app**:
model → service → Test or UI
- ✓ Build **reusable** and **interactive** Flutter UIs
- ✓ Build **dynamic UI**: connect data with UI
- ✓ Implement **navigation** (Navigator, GoRouter)

***Values and dispositions** you need to develop*



BEHAVIOR

- ✓ **Own and defend** your technical **decisions**
- ✓ Take **responsibility** for your own homework habits
- ✓ **Learn** from **feedback** and keep **trying hard**
- ✓ Use **AI responsibly** without **losing coding autonomy**

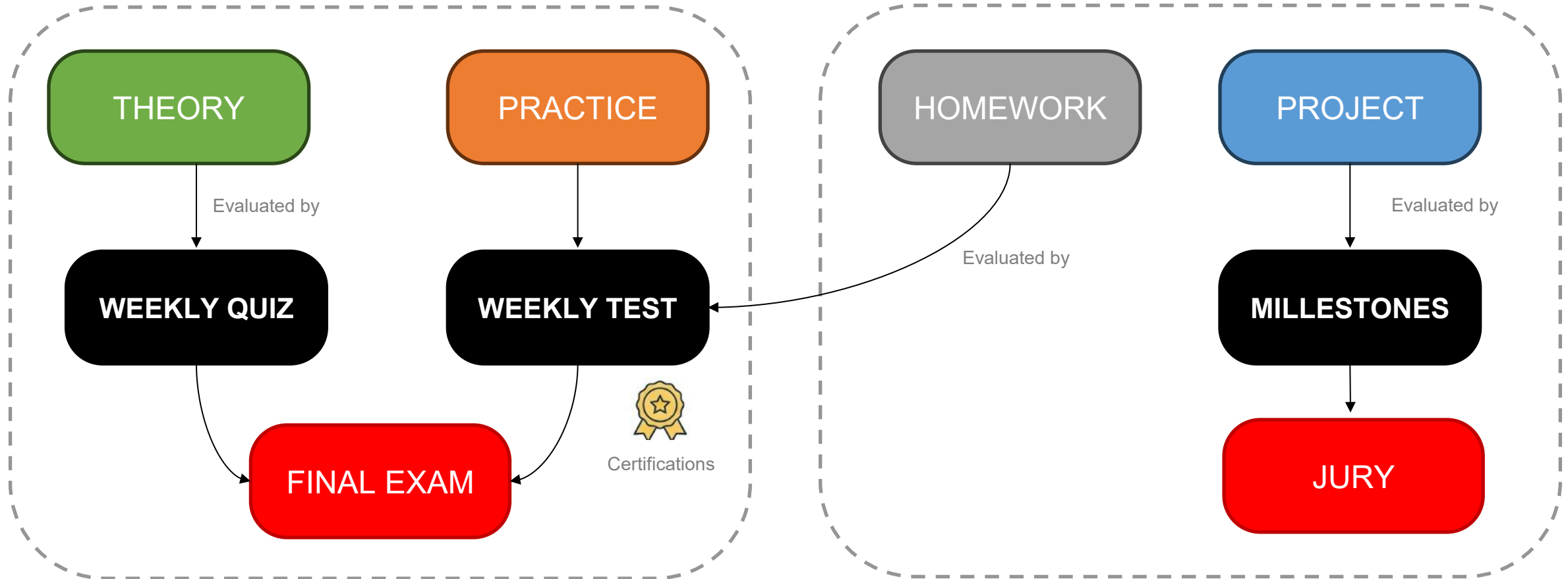
Course Agenda

	DATE	THEORY	PRACTICE	COURSE PROJECT	CERTIF/EXAM
W1	28 SEP	Dart Basics			
W2	05 OCT	Dart OOP			
W3	12 OCT	Architecture & Tests		Business Meeting	
W4	19 OCT	Flutter Basics			
W5	26 OCT				Modeling Certification
W6	02 NOV	Stateless Widgets		Modeling Meeting	
W7	09 NOV	Layouting			
W8	16 NOV	Stateful Widgets			Static UI Certification
W9	23 NOV	Loops, Conditions, Callbacks		Static UI Meeting	
W 10	30 NOV	List View & Modals			
W 11	07 DEC	Inputs & Forms			Dynamic UI Certification
W 12	14 DEC	Navigation		Dynamic UI Meeting	
W13	21 DEC				PRE EXAM
W14	28 DEC			Jury	FINAL EXAM

How will you **learn** during this course?

IN - CLASS

OUT OF CLASS



Be ready for **Theory** sessions

BEFORE

✓ Study the concepts deeply

Read the course material, ask AI, read documents, take notes using your own words

✓ Test yourself

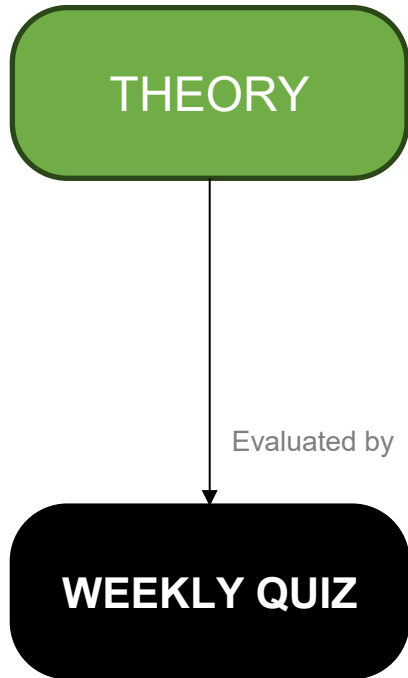
Ask AI to generate a quiz – given the week objectives and evaluate your answers

DURING

✓ Every theory session includes a **10-minute MCQ quiz**

The principle is simple: if you have understood the concepts, you should pass this quiz!

✓ Don't be late ! The quiz start on time 😊



Be ready for **Practice** sessions

BEFORE

- ✓ Perform the homework **honestly**

Homework is to train yourself, not a proof of your skill achievement

- ✓ We don't assess homework !

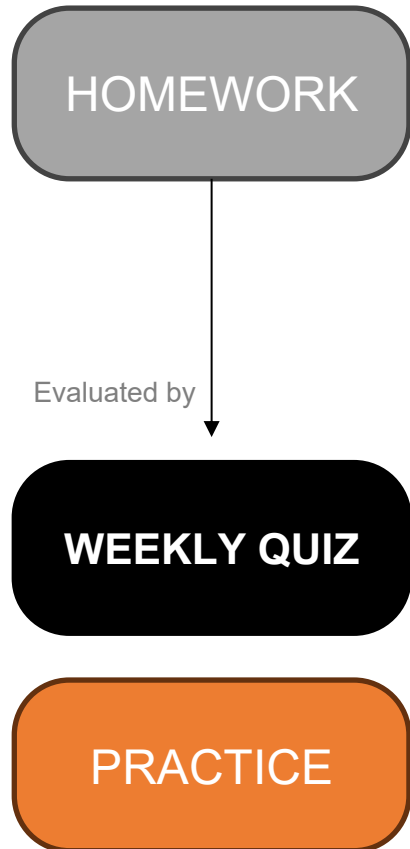
*We assess what you can do **after homework***

DURING

- ✓ Every practice session starts with a **30-minute test**

The principle is simple: if you have completed the homework, you should pass this test!

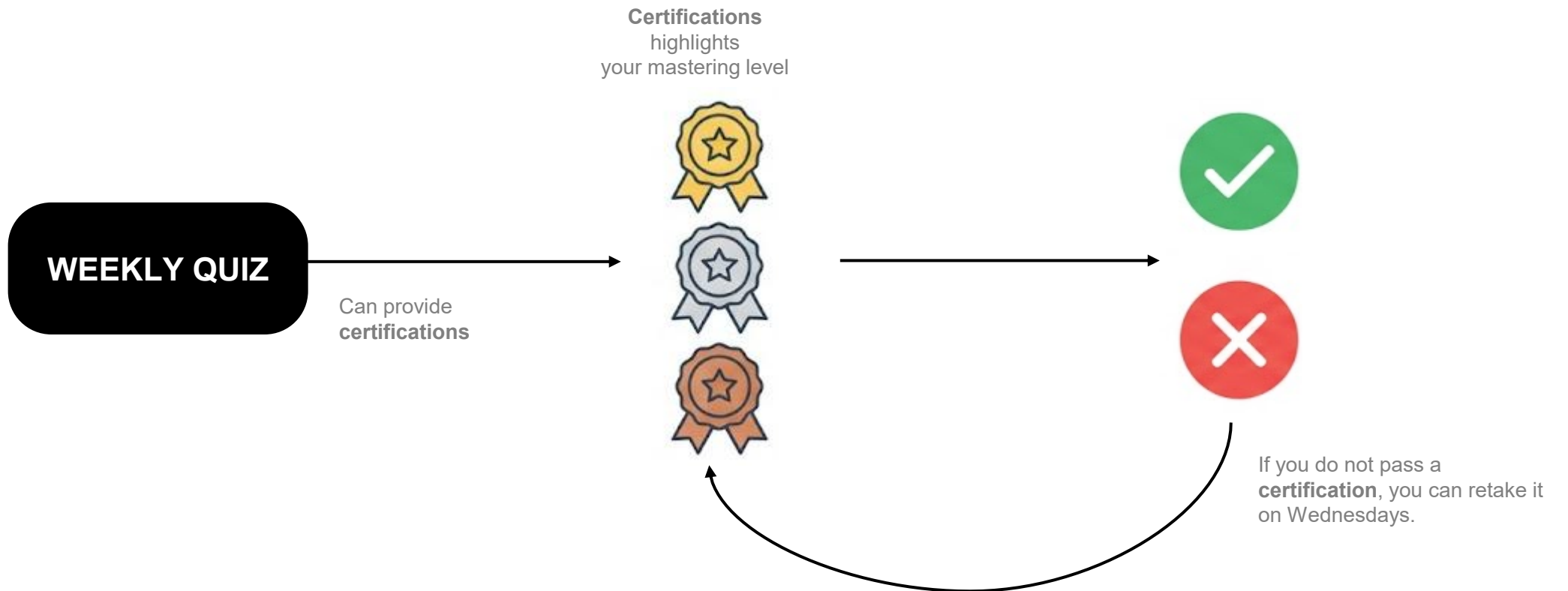
- ✓ Come in time, since the test start on time 😊



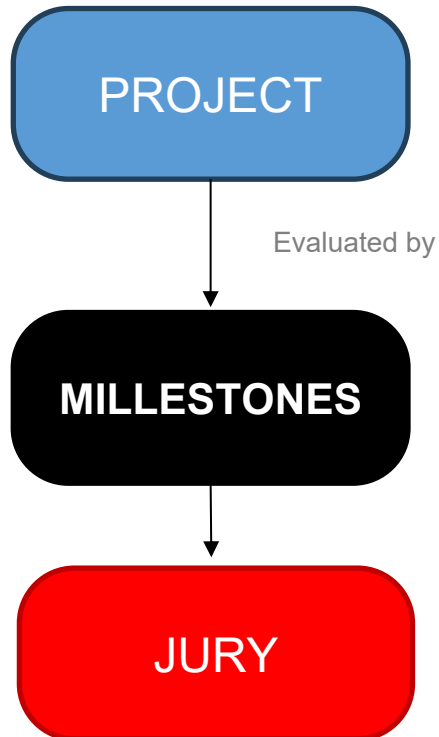


Certifications

- ✓ Some of the practice tests will become **certifications**
- ✓ If you do not pass a **certification**, you can retake it on Wednesdays.



Be ready for the **Embedded Project**



- ✓ The Project is developed **throughout the course**

Project start form WEEK 2

- ✓ Throughout the course, you will have **milestone** interviews:

We assess the process: thinking, reasoning, engineering decisions throughout the project —not just the final product !

- ✓ The Project is done in **pair** or **individually**

GRADING POLICY

The **final course grade** is calculated using the following weighted components:

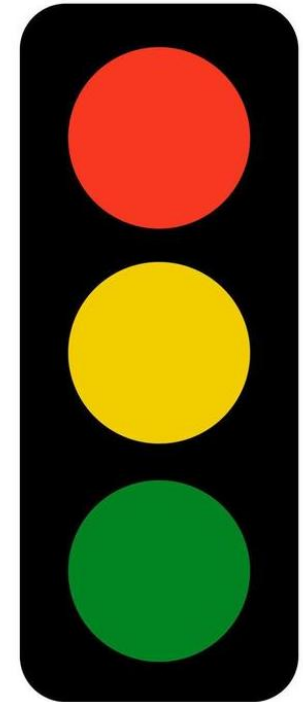
TYPE OF EVALUATION	WEIGHT
Attendance & Participation	10 %
Theory Quizzes	10 %
Practice Tests	20 %
Course-Embedded Project	20 %
Final Exam	40 %
Final Course Grade	100 %

More than 3 unauthorized absences will result in a score of 0 for the course.

- If the final course grade is below 50, the student must take a retake exam
- If the retake exam score is 50 or above, the student's final course grade is adjusted to 50 (Grade C)
- If the retake exam score is below 50, the student fails the course and must take the course again

AI Policy

- Exam and tests are **LEVEL RED**
 - AI tools are strictly prohibited.
 - AI use is treated as academic misconduct
- Homework and projects are **LEVEL YELLOW**
 - AI tools allowed for **research**, explanations
 - AI tool **forbidden** to produce **code**



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AI FRAMEWORK



Seniors already know **architecture**, **SOLID** principles, they master **coding**, **debugging**...

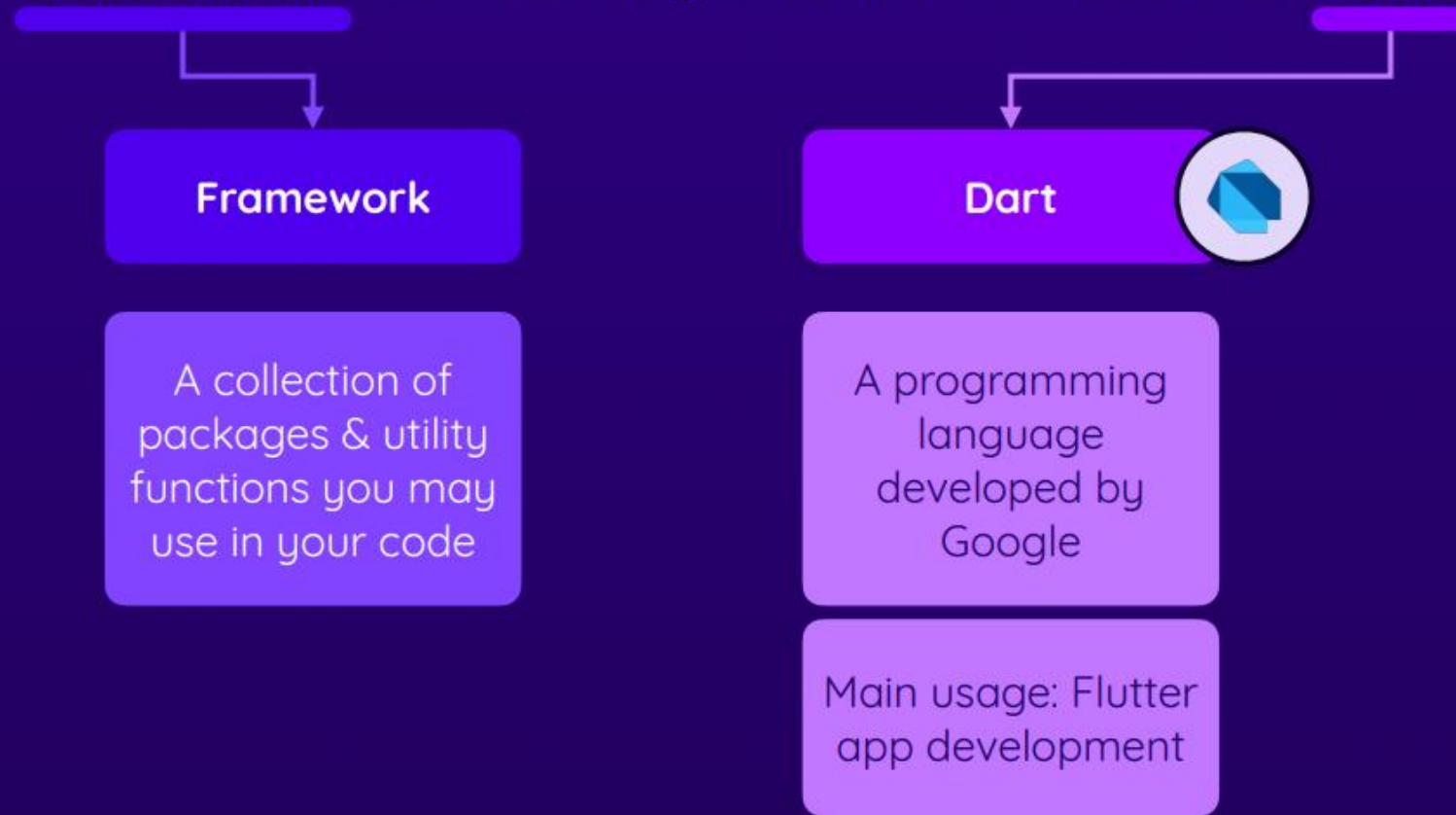




Learning is a process
made of detours, mistakes

Flutter is **not** a programming language!

It's a **framework** for building user interfaces with **Dart**





Dart





Language ^

Introduction

Variables

Operators

Comments

Types ^

Built-in types

Records

Collections

Generics

Typedefs

Type system

Patterns v

Control flow ^

Loops

Branches

Error handling

Functions

Language > Variables

Variables

Learn about variables in Dart.

Here's an example of creating a variable and initializing it:

```
var name = 'Bob';
```

Variables store references. The variable called `name`

The type of the `name` variable is inferred to be `String`. If you want to restrict the variable to a single type, specify the `Object` type:

```
Object name = 'Bob';
```

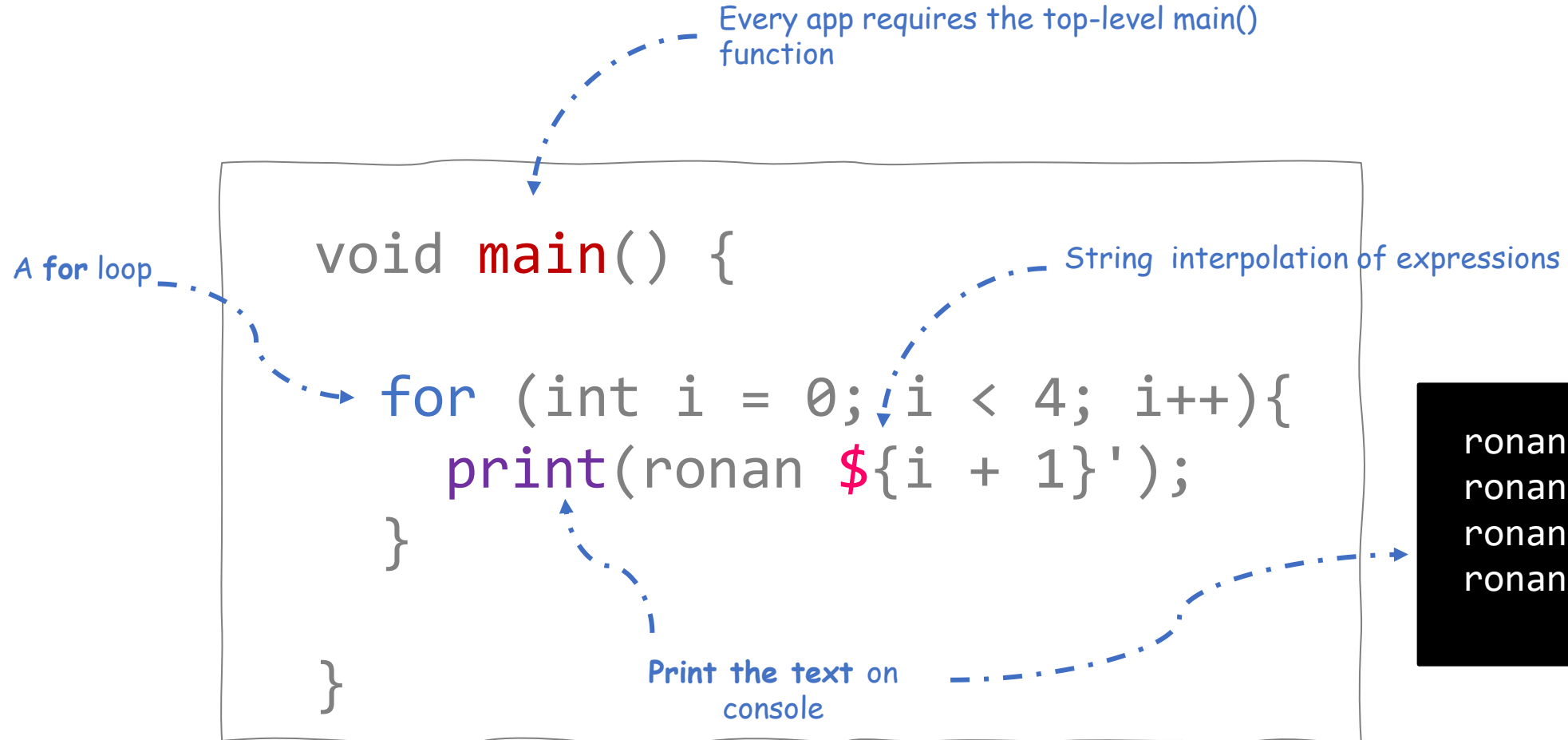
Another option is to explicitly declare the type of the variable:

```
String name = 'Bob';
```

DOC

A First DART Program !

DOC



```
ronan 1  
ronan 2  
ronan 3  
ronan 4
```

Try it on console : <https://dartpad.dev/?theme=dark>

Dart Variables

DOC

In Dart, we can either **specify** or **infer** a variable type

SPECIFY TYPE

```
String name = 'Bob';
```

INFER TYPE

```
var name = 'Bob';
```



The type of the name variable is inferred to be String

Dart 'guess' the type by checking the value assigned to the variable

Dart Built-in Types



Dart comes with some built-in types.

Type	Description	Example
int	Store an integer	4578 5 -8784
double	Store a decimal	3.14 -1.5
String	Store a text	“double quotes” ‘single quotes’;
bool	Store a Boolean	true false 4>5
List	Store an array of elements	List<int> scores = [1,2,3]; var scores = [1, 2, 3];

← Inferred type

Dart Loops



You can loop either with the **standard for loop**, or using **for-in** with an iterable structure

STANDARD FOR LOOP

C - like for loop

```
for (int i = 0; i < 5; i++) {  
    print(i);  
}
```

Print on console

FOR-IN ON ITERABLE

```
List<int> numbers = [1,2,3];
```

Numbers must be iterable

```
for(int num in numbers) {  
    print("Number is $num");  
}
```

Dart Functions



Dart is a true object-oriented language, so even functions are objects and have a type

DEFINE A FUNCTION

A function can
return a value

A function
need a name

A function can
have parameters

```
bool isEven(int number) {  
    return number%2==0;  
}
```

A function must be
defined before using it

CALL A FUNCTION

Function call
With the needed
parameters

```
void main() {  
    bool result = isEven(4);  
    print(result);  
}
```

true

What will be the output?

```
int compute(List<int> numbers) {  
    int sum = 0;  
  
    for (var n in numbers) {  
        if (n % 2 == 0) {  
            sum += n;  
        }  
    }  
  
    return sum;  
}  
  
void main() {  
    List<int> values = [1, 2, 3, 4, 5];  
    print(compute(values));  
}
```

- A. 6
- B. 9
- C. 12
- D. 15

ANSWER

What will be the output?

```
int compute(List<int> numbers) {  
    int sum = 0;  
  
    for (var n in numbers) {  
        if (n % 2 == 0) {  
            sum += n;  
        }  
    }  
  
    return sum;  
}  
  
void main() {  
    List<int> values = [1, 2, 3, 4, 5];  
    print(compute(values));  
}
```

- A. 6
- B. 9
- C. 12
- D. 15



Create your **first function** in Dart

Write a function called:

```
void printStudentInfo(String name, int age, List<int> scores, bool isCambodian)
```

This function should:

- ✓ Print the student's name
- ✓ Print the student's age
- ✓ Print whether the student is Cambodian (Yes or No)
- ✓ Print all the scores using a for-in loop

THX YOU !



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Flutter

RESSOURCES FOR THIS COURSE

FLUTTER DOC

Dart, Flutter, Pub Official websites

<https://dart.dev/>

<https://flutter.dev/>

<https://pub.dev/>

Flutter YouTube channel

<https://www.youtube.com/@flutterdev>



Flutter Cookbook

<https://docs.flutter.dev/cookbook>

FLUTTER CLEAN CODE GUIDE

[See this guide](#)

FLUTTER MOOCS

[Udemy - Dart & Flutter Course](#)

[ClassCentral MOOC](#)

[Udemy - Figma - UI UX](#)



IDE / TOOLS – *if needed*

Online Flutter editor

<https://zapp.run/profile>

Online Dart editor

<https://dartpad.dev/>

